

Amandine Brunetto

RESEARCH SCIENTIST

☎ +33 6 30 01 88 55 | ✉ amandine.brunetto@gmail.com | 🏠 amandinebttto.github.io | 📺 AmandineBtto | 🌐 amandine-brunetto

Multimodal AI researcher specialized in audio-visual generative models and cross-modal learning, with publications at CVPR, ICLR, and ICML.

Work Experience

Postdoctoral Researcher

IMAGINE, ENPC, IP Paris

January 2026 - now

- Finetuning a body model with LoRA to improve 3D hand motion estimation from low-resolution sign-language videos.
- Supervise PhD students on research direction, experimental design, and methodology.

AI Research Scientist

Mines Paris - PSL

November 2022 – October 2025

- Pioneered a flow matching model based on a diffusion transformer for one-shot acoustic scene generation from multimodal context. Created novel scene-consistency metrics based on a joint-embedding model between sight and sound (CVPR 2026).
- Introduced a cross-modal learning framework that integrates neural radiance fields (NeRF) with neural acoustic fields for real-time novel-view audio-visual synthesis (ICLR 2025).
- Conducted large-scale experimentation and optimization on HPC infrastructure.
- Released all research code and datasets as open source.

Computer Vision Engineer

Mines Paris - PSL

May 2022 – September 2022

- Built a robot and developed a complete data recording pipeline to capture multimodal audiovisual data. Processed the data to create a large-scale audio-visual dataset, which was released to the robotics community (IROS 2023).
- Developed monocular depth estimation models using audio cues to boost accuracy in complex scenes.

Research Intern

Mines Paris - PSL

May 2021 – July 2021

- Developed unsupervised deep learning methods for video-based classification of traffic situations, incorporating object detection, classification, and tracking to enhance scene understanding.

Education

Ph.D. in Artificial Intelligence

PSL University

2022 – 2025

Paris, France

- Research topic: "Combining Sight and Sound: Multimodal Reasoning for Geometric and Acoustic Scene Reconstruction".

M.Sc in Technology and Innovation Management

Paris Dauphine - PSL University

2021 – 2022

Paris, France

- Graduated first of the class and with the highest honors.
- Collaborated on diverse interdisciplinary projects, developing creative and innovative solutions to complex challenges.

M.Sc in Engineering

ENSEA

2019 – 2022

Cergy, France

- Specialized in signal processing and computer science.

Publications

CVPR 2026

A. Brunetto. *Few-shot Acoustic Synthesis with Multimodal Flow Matching*

ICLR 2025

A. Brunetto, S. Hornauer, F. Moutarde. *NeRAF: 3D Scene Infused Neural Radiance and Acoustic Fields*

ICML 2025

G. Kasmi, **A. Brunetto,** T. Fel, J. Parekh. *One Wave to Explain them All: A Unifying Perspective on Post-hoc Explainability*

IEEE IROS 2023

A. Brunetto, S. Hornauer, S. Yu, F. Moutarde. *The Audio-Visual BatVision Dataset for Research on Sight and Sound*

Skills

AI: Multimodal learning, generative models, transformers, computer vision, audio, signal processing, NeRF, human motion estimation

Programming: Python, PyTorch, Lightning, Librosa, Torchaudio, Git, SLURM, Linux

Robotics: ROS, embedded systems, dataset creation

Soft skills: Multidisciplinary collaborative work, project management, popularization of complex topics, creative problem-solving

Personal Interests

Classically trained musician (piano and flute), with conservatoire experience. I also explore art history as part of a broader interest in culture and visual expression.